SMART GRIDS / PROJECT REPORT

Prosumer community simulator with battery storage

Implementation and reproducible experiment results
21 October 2026

A Python simulation of **100 households in 10 neighborhoods**, all equipped with rooftop PV. Exactly **55 households own batteries**. The simulator compares neighborhood electricity trading, a community aggregator and external-grid settlement, with and without price guidance and funded seller rewards.

Research objective

Encourage peer-to-peer (P2P) trading while conserving energy and accounting for every payment. Measure how storage changes both the amount traded and the surplus available to trade. Record every completed internal-market trade on a simplified Proof-of-Work blockchain.

Scenario	Battery operation	Regulation	Charging priority
A	55 owners	Disabled	Self-consumption first
B	55 owners	Enabled	Self-consumption first
C	Disabled	Disabled	Counterfactual
D	Disabled	Enabled	Counterfactual
E	55 owners	Disabled	P2P before charging
F	55 owners	Enabled	P2P before charging

Observed outcome

With batteries, regulation increased P2P energy from **8.935 to 18.656 kWh**. The P2P share increased from 0.626% to 1.308%. However, household net cost increased because collected fees exceeded rewards. These are separate outcomes, not a claim that regulation makes every participant better off.

Evidence boundary: The default weather, demand and price history are synthetic. One deterministic 24-hour experiment demonstrates implementation behavior; it does not establish real-world or statistically general performance. All six scenarios use identical hourly inputs and fixed ownership.

02 Community layout and household types

The community is shown as a **10 x 10 square schematic**. Each row is one trading neighborhood, with ten households. IDs, PV capacities and ownership are taken from the same household configuration used by the simulator.



Figure 1. Actual household configuration. Blue homes have PV plus a battery (B marker); gold homes have PV only. All 100 homes have PV. The 55 battery owners retain their identity in every scenario.

Groups G1, G2, G5, G7 and G10 have six battery owners each; G3, G4, G6, G8 and G9 have five each. Assignment uses fixed seed 42. Positions are display choices, not geographic coordinates, feeder topology or distance-based trading constraints.

Automatic output: a normal simulation run generates this map as PNG and SVG, together with cell and neighborhood CSV tables. The **--no-charts** option deliberately skips image generation. C/D disable battery operation while retaining the community ownership design.

03 Inputs and physical energy model

Community and forecasts

Household PV capacity ranges from 2 to 8 kW. Daily household demand ranges from 8 to 24 kWh, with individual morning and evening peaks, shifted timing and seeded variation. Battery ownership is distributed as five or six owners per neighborhood using seed 42. Capacities, profiles and offer parameters are retained across scenarios.

Synthetic irradiance follows a daylight curve between 06:00 and 20:00, with a 900 W/m2 peak and cloud factors from 0.65 to 1. The illustrative simulation day starts on 15 June 2026 at UTC+02:00. This date is an input setting, not a weather observation.

PV energy (kWh) = installed capacity (kW) x irradiance / 1000 x performance factor x step hours. The default performance factor is 0.85. PV output is zero at night and capped at installed capacity times the step duration.

Reference electricity prices use historical hourly averages from observations strictly before the experiment start. The default history contains 14 synthetic days. Optional CSV adapters accept weather forecasts, household demand forecasts and historical prices. Missing coverage, duplicate keys and non-finite inputs are rejected. Weather and demand CSV provenance must establish that the forecasts were available when issued.

Battery defaults and constraints

Parameter	Starting value	Interpretation
Capacity / power	10 kWh / 3 kW	Charge and discharge limits
Efficiency	95% each way	Losses explicitly accounted for
SOC bounds	10% to 90%	Initial SOC 20%
Charging source	Owner PV only	No grid or P2P charging

These battery values are suggested engineering defaults. Each battery stores its energy state across hours. Charging and discharging are measured on the household AC side. Stored energy evolves as:

```
E_next = E_start + eta_c * charge - discharge / eta_d
```

PV serves immediate demand first. A battery may discharge only to cover its owner's remaining demand, and may charge only from its owner's remaining PV surplus. It cannot charge and discharge simultaneously or export stored energy. A/B charge before P2P; E/F attempt P2P sales before storing unsold PV, then proceed to the aggregator.

Household energy identity:
PV + discharge + purchases = demand + charge + sales.
Community identity:
PV + grid imports + initial storage = demand + grid exports + final storage + battery losses.

04 Markets, regulation and blockchain

Hourly market sequence

1. Self-balancing. Calculate household demand served by PV and eligible battery flows.
2. Neighborhood P2P. Shuffle buyers and seller encounter orders. Each buyer may try every seller in its group. Compatible bids execute at the seller's ask, limited to both parties' remaining energy.
3. Community aggregator. Match lowest asks with highest bids across groups. Execute compatible trades at the bid-ask midpoint.
4. External grid. Import remaining deficits at EUR 0.30/kWh and export remaining surplus at EUR 0.06/kWh.

Price guidance and rewards

Original offers multiply the price forecast by household-specific parameters and respect price bounds. Regulation blends offers toward 1.04 times the reference for bids and 0.96 times the reference for asks, with strength 0.65. Guidance affects P2P only; aggregator offers return to their original values.

Opening regulator funds	P2P seller reward
Below EUR 1	EUR 0/kWh
EUR 1 to below EUR 5	EUR 0.01/kWh
EUR 5 or more	EUR 0.02/kWh

The reward tier is fixed at the start of the hour. Before a discounted proposal is guaranteed, the regulator reserves the full reward for its feasible quantity. The adjusted ask equals the guided ask minus 50% of the reward rate. A successful seller receives the adjusted trading payment plus the full reward.

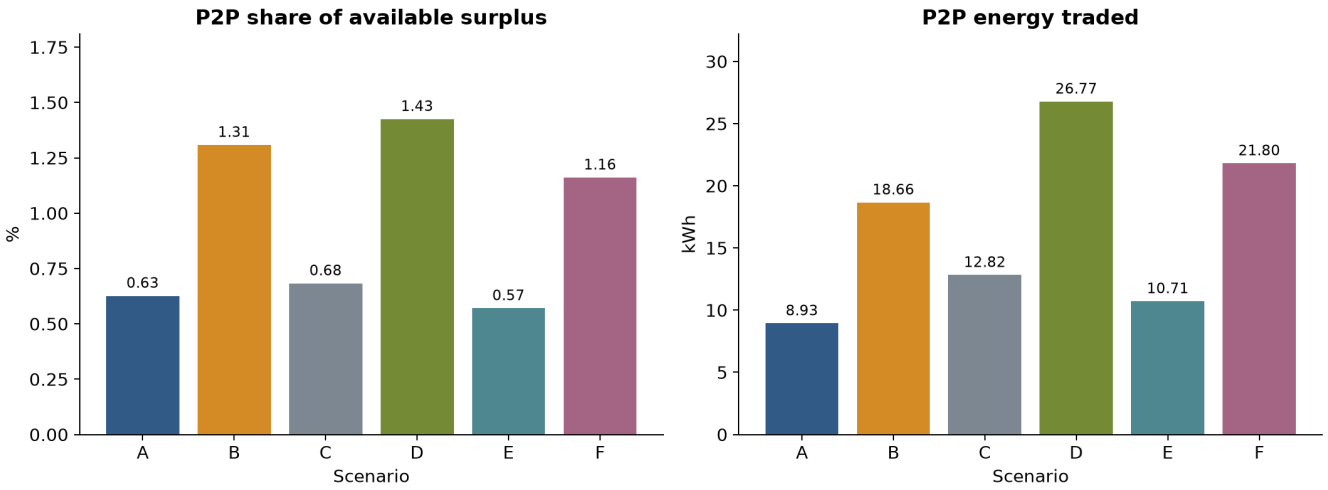
Rejected quotes release reservations. If the reward is unaffordable, matching tries the original offers without a subsidy. There are no partial-quantity subsidies. The fund starts at EUR 0. Regulated grid imports and exports each pay EUR 0.01/kWh after trading; those new fees can finance rewards only in later hours. Internal trades carry no regulator fee.

Independent blockchain record

Ten simulated miners take turns searching SHA-256 nonces inside one Python process. Exactly one block is mined per hour, including hours with no internal trades. Every block hash begins with **000** and references the preceding hash. The first block references 64 zeros; no extra genesis block is added.

All completed P2P and aggregator trades appear exactly once on-chain. Validation checks contents, unique transaction IDs, proof of work, block linkage and agreement with the trade journal. Grid exchanges and every cash movement remain in separate ledgers. Mining does not choose electricity quantities or prices.

05 P2P experiment results



Identical hourly inputs across scenarios. See summary.json for data sources and seed. A/B: 55 batteries. C/D: none. E/F: P2P priority. B/D/F: regulated.

Figure 2. Actual P2P outcomes for the shared synthetic day. A/B have batteries; C/D disable them; E/F use P2P-priority charging. B/D/F are regulated.

Scenario	P2P kWh	Share %	Available surplus kWh	P2P trades
A	8.935	0.626	1,426.165	38
B	18.656	1.308	1,426.165	79
C	12.822	0.683	1,878.065	62
D	26.766	1.425	1,878.065	124
E	10.709	0.570	1,878.065	51
F	21.801	1.161	1,878.065	104

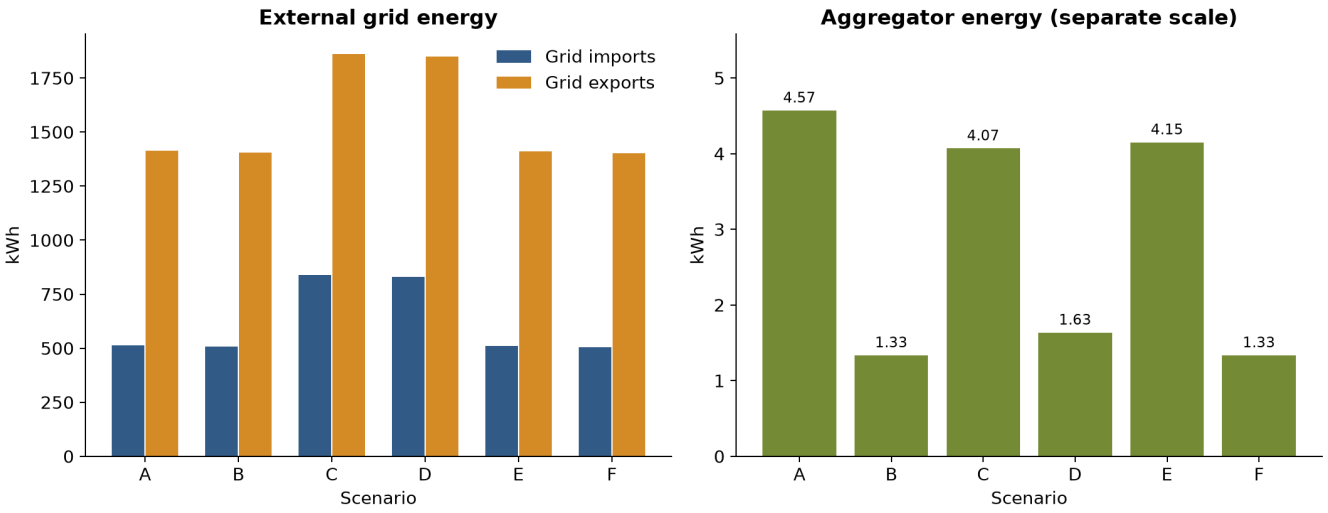
KPI definition and interpretation

P2P share (%) = 100 x total P2P energy / total surplus available after self-balancing. For P2P-priority storage, the denominator is the surplus offered before P2P and before deferred charging. A zero denominator is reported as undefined (JSON null).

Regulation increased P2P energy by 9.721 kWh in B versus A, 13.943 kWh in D versus C, and 11.091 kWh in F versus E. The heuristic can improve price compatibility but cannot create surplus, demand or a globally optimal trade allocation.

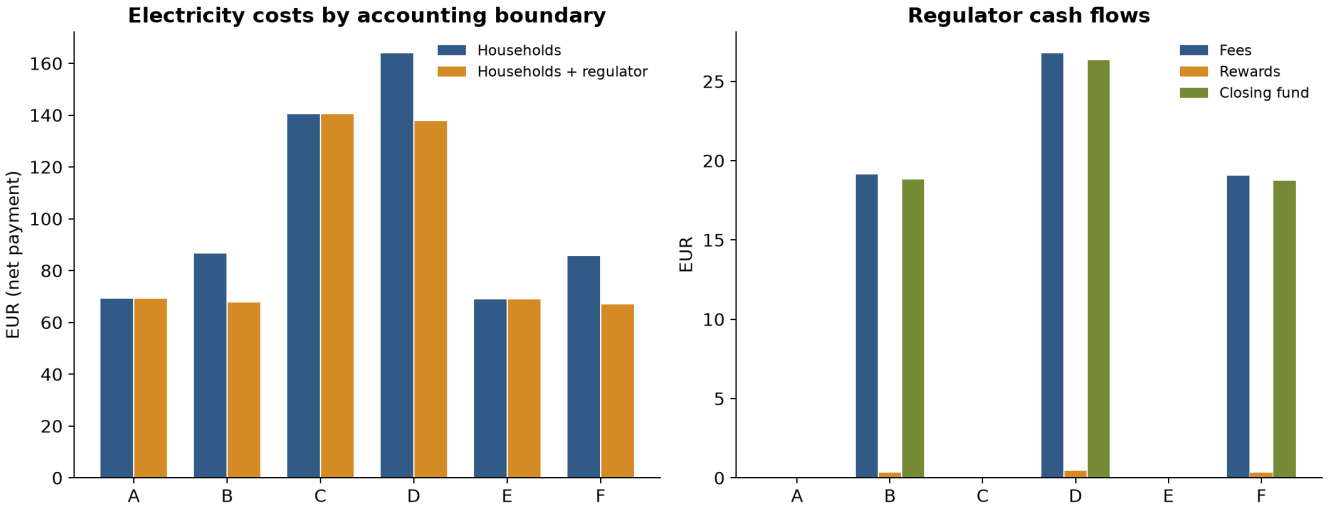
E trades more P2P energy than A but has a lower percentage share because its denominator is larger. This illustrates why absolute energy, P2P share and raw PV surplus must be assessed together. Low shares in this particular setup should not be extrapolated to another community or season.

06 Grid exchanges and electricity costs



Identical hourly inputs across scenarios. See summary.json for data sources and seed. A/B: 55 batteries. C/D: none. E/F: P2P priority. B/D/F: regulated.

Figure 3. Grid imports/exports and aggregator energy. The aggregator uses a separate scale because its volumes are much smaller than external-grid volumes.



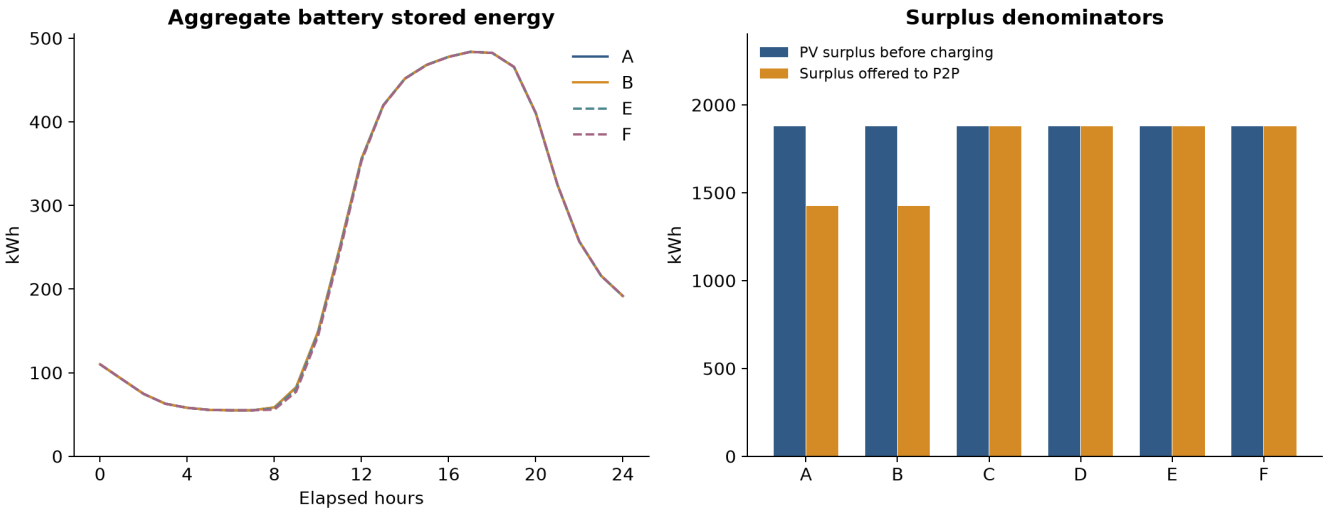
Identical hourly inputs across scenarios. See summary.json for data sources and seed. A/B: 55 batteries. C/D: none. E/F: P2P priority. B/D/F: regulated.

Figure 4. Household net payments versus payments for households plus the regulator, and regulator fees, rewards and closing funds.

In A/B, imports fall from 513.209 to 506.723 kWh and exports from 1,412.661 to 1,406.176 kWh. Household net cost rises from EUR 69.20 to EUR 86.47, while household-plus-regulator cost falls from EUR 69.20 to EUR 67.65.

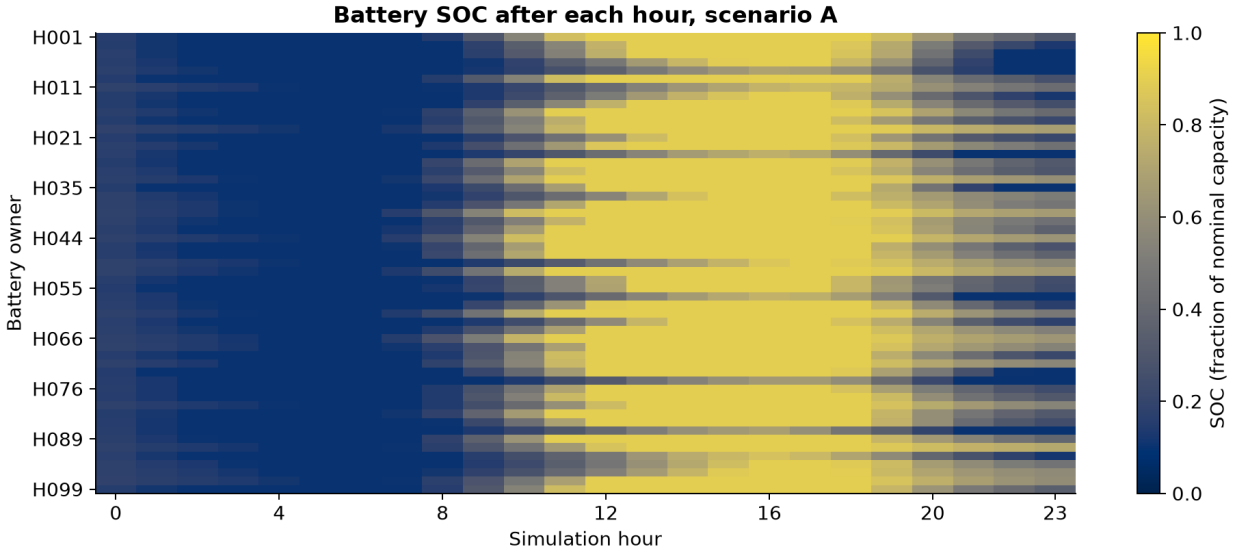
The difference is the regulator's retained funds. Internal payments transfer money within the community; they are not an external resource cost. Costs exclude battery investment, degradation, taxes and inventory valuation.

07 Battery operation and surplus availability



Identical hourly inputs across scenarios. See summary.json for data sources and seed. A/B: 55 batteries. C/D: none. E/F: P2P priority. B/D/F: regulated.

Figure 5. Aggregate stored energy and the surplus denominators. Similar storage trajectories overlap. Raw PV surplus is shared across all scenarios; charging reduces the surplus offered in A/B.



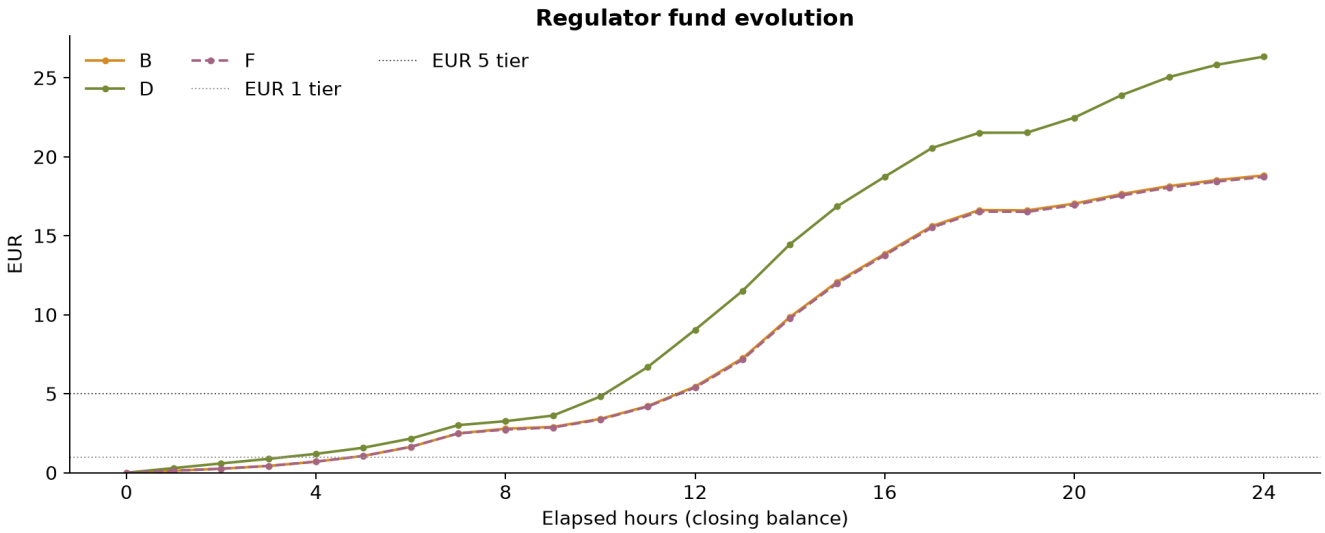
Identical hourly inputs across scenarios. See summary.json for data sources and seed. A/B: 55 batteries. C/D: none. E/F: P2P priority. B/D/F: regulated.

Figure 6. End-of-hour SOC for all 55 owners in scenario A. Color shows a fraction of nominal battery capacity; allowed operating bounds are 0.10 to 0.90.

Across the community, A/B charge 451.899 kWh, discharge 330.274 kWh and lose 39.978 kWh through conversion. Aggregate initial storage is 110.000 kWh; final storage is 191.648 kWh.

Raw PV surplus before charging is 1,878.065 kWh in every scenario. E/F finish with 191.628 kWh stored. Compare those inventories when interpreting electricity costs. C/D retain ownership metadata but have zero battery flows and zero active stored energy.

08 Regulator fund and financial conservation



Identical hourly inputs across scenarios. See summary.json for data sources and seed. A/B: 55 batteries. C/D: none. E/F: P2P priority. B/D/F: regulated.

Figure 7. Closing regulator funds through the day, with EUR 1 and EUR 5 reward thresholds. Each hour's reward rate uses opening funds, not the plotted end-of-hour fees.

Scenario	Grid fees EUR	Rewards EUR	Closing fund EUR
B	19.129	0.310	18.819
D	26.783	0.441	26.342
F	19.066	0.341	18.725

Accounting boundary

Household net cost = electricity purchases - sale revenue + regulator fees - seller rewards. Internal P2P and aggregator payments cancel when all households are combined. Every cash movement has one payer and one payee.

```
Fund_close = Fund_open + grid_fees - P2P_rewards
Community_cost = Household_cost - change_in_fund
```

The second expression includes both households and the regulator and equals net external-grid payments. Fund accumulation therefore increases households' net payments without creating or destroying money. Household accounts have no starting-cash spending restriction; the regulator alone is constrained by available reward funding.

Rewards are small relative to collected fees in this run. The current rule does not return the remaining fund at day end. A future experiment could study a different fee or reward policy, but those changes would require explicit assumptions and a new comparison.

09 Verification, reproduction and limits

Recorded execution

The implementation was exercised on Python 3.14.7 on macOS arm64. The final automated suite completed with **59 passing tests**. All six scenarios completed 100-household, 24-hour runs with 24 valid hourly blocks each. These results come from executed code, not estimated outputs.

Tests cover household composition, forecast cutoff and CSV adapters, SOC/power/loss constraints, energy conservation, price and quantity rules, funded rewards, fee timing, blockchain tampering and coverage, KPI denominators and exported-file integrity. The independent verifier reopens the saved files and checks the input hash, trade journals, blockchains, household energy and cash movements.

Maximum absolute energy reconciliation error: 2.27e-13 kWh.
Maximum absolute financial reconciliation error: 1.42e-14 EUR. These are numerical floating-point residuals.

Reproduce from the project directory

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m pytest -q
python main.py --include-priority
python scripts/verify_results.py
```

Use **config/default.json** or a partial JSON override with **--config** to change parameters. The README documents command-line options and all exported files. CSV input formats and forecast provenance requirements are in **docs/input_formats.md**.

Limitations and conclusion

The model excludes feeder capacity, voltage, network power flow, battery aging, investment cost and strategic learning. It uses hourly energy settlement, a fixed UTC offset and illustrative daily demand profiles. Synthetic data is not calibrated to a measured community. Other Python versions were not executed here.

For this input set, regulation increases direct P2P trading while raising household net payments through retained fees. Batteries reduce grid dependence and also reduce the surplus offered for trading. P2P-priority charging highlights the distinction between traded energy and its percentage denominator. These findings describe this demonstration, not a guaranteed improvement under every input condition.

Local source record

Results: results/comparison.json, scenario summary.json files and charts.
Validation: docs/validation.md, docs/execution.log, docs/export_verification.json.
Specification and modeling choices: supplied project brief, README.md and config/default.json.

Input SHA-256: c5537f8e26fb4e01d9157db48b5ae8dc03c5f5faa4cb6efb72e44bc493c71d31